In 2018 the share of renewable materials of all raw materials used across Huhtamaki was 67% (66%).

Most of the renewable materials were wood fiber-based materials, either virgin fibers or recycled paper. The most used recycled raw material was post-consumer recycled paper. Virtually all virgin fiber used by Huhtamaki has PEFC™, FSC® or SFI® Chain of Custody certification, which guarantees that the fiber is traceable to sustainably managed forests.

The share of recycled materials in 2018 remained essentially stable at 27% (27%).

Energy

302-1 Energy consumption within the organization

We use both primary energy sources (fuel) and secondary energy (electricity and heat) in our operations.

Our total energy consumption in 2018 remained essentially flat compared to 2017 at 2,182 GWh (2,177 GWh). Consumption of primary energy, which is mainly used in the drying process of recycled fiber-based packaging, was 991 GWh (996 GWh). Consumption of secondary energy in 2018 was 1,191 GWh (1,181 GWh). The main source of secondary energy was electricity, which accounted for 99% (98%) of the total consumption. The share of renewable energy remained stable year-on-year at 3%.

302-3 Energy intensity
302-4 Reduction of energy consumption

We are continuously looking for ways to reduce energy use and improve efficiency, for the benefit of our business and the environment. This work is mainly driven by the Lean Six Sigma community within Huhtamaki, who identify and address opportunities to save energy. In addition, we are working on the Natural Resource Plan, a collaborative project to identify the natural resource reduction opportunities across our entire business. Energy is an important part of this project, which will be completed in 2019.

In 2018, our energy efficiency continued to improve, by 2% compared to 2017. Energy consumption per sellable ton produced (STP) was 2.04 MWh/STP (2.08 MWh/STP).

As a result of the energy efficiency improvements, we have reduced energy consumption by 21 GWh compared to 2017 when corrected for the production volume. Reductions are typically achieved through operational excellence projects. These numbers include all forms of energy used within Huhtamaki.

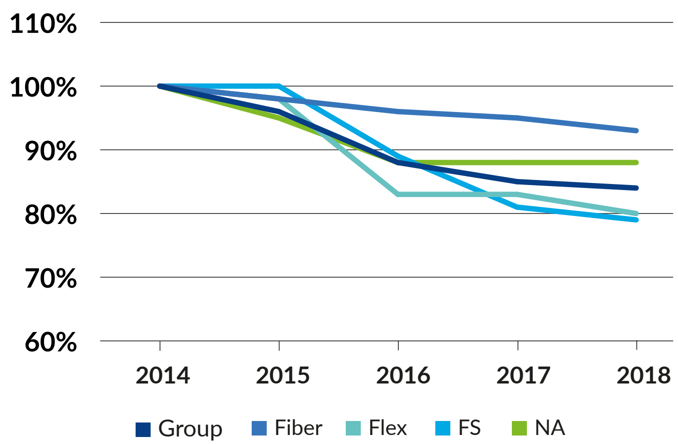
Our energy efficiency in relation to production output has improved by 16% over the past five years.

Energy consumption (GWh and %)

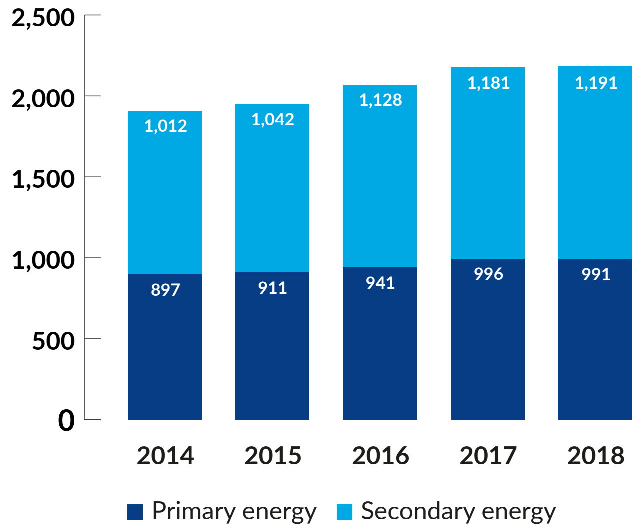
		2018		2017		2016	
Primary energy consumption	Gas	912	42%	923	43%	849	41%
	Oil	79	3%	72	3%	80	4%
	Coal	0	0%	0	0%	11	1%
	Renewable fuels	0	0%	0	0%	0	0%
Secondary energy consumption	Electricity	1,176	54%	1,162	53%	1,111	53%
	Share of renewable electricity	69	6%	62	5%	-	-
	Other secondary energy	15	1%	19	1%	17	1%
Total energy consumption		2,182	100%	2,177	1%	2,069	100%

Other secondary energy: heating, cooling and steam combined due to small consumption. No energy sold. Units report energy consumption in a specific management system.

Indexed energy efficiency (MWh/STP)



Total energy consumption (GWh)



Water

303-1 Water withdrawal by source

In our operations, water is mainly used in the manufacture of fiber packaging and as cooling water. The amount of water used in absolute terms is limited, but due to scarce water resources globally, we consider water as a material aspect in our Packaging for Good program.

Our fresh water intake in 2018 increased to 9.5 million m³ (9.1 million m³), driven by an increase of surface water intake due to relatively hot weather in the United States requiring more cooling water for certain units. The water efficiency of the Group somewhat decreased as water intake in relation to manufacturing output increased by 3%. Half of our water intake is surface water. The most water-intensive business at Huhtamaki is rough molded fiber packaging manufacturing; however, many of these manufacturing units have efficient water recirculation systems in place.

Our manufacturing units are responsible for working together with stakeholders to address any water related issues. For example, our fiber manufacturing unit in South Africa engaged with local authorities and communities during the drought in the Cape Town area and acted on this by reducing our withdrawal of surface water.

Huhtamaki's own KPI: All plants in water stressed areas have a water management plan

We have used the World Resources Institute's Aqueduct water risk mapping tool to investigate the possible water risk at our manufacturing units. Three of our manufacturing units are in areas of "highest physical risk". However, when corrected for the relatively low volume of water consumed by these three units, their overall water risk is low to medium. Fresh water scarcity is an important topic to address and water is also a material topic for Huhtamaki. Therefore, water management plans for these manufacturing units are important and we will report on the progress in our next corporate responsibility report.

Water withdrawal by source (million m³ and % of total)

	2018		2017		2016	
Surface water	4.8	50%	4.5	49%	4.4	49%
Ground water	1.2	13%	1.1	13%	1.2	13%
Municipal water supply	3.5	37%	3.5	38%	3.4	38%
Total	9.5	100%	9.1	100%	9.0	100%

Emissions

305-1 Direct (Scope 1) GHG emissions
305-2 Energy indirect (Scope 2) GHG emissions
305-4 GHG emission intensity
305-5 Reduction of GHG emissions

Greenhouse gas (GHG) emissions are one of the main negative environmental impacts of our manufacturing operations. We continuously work to improve our energy efficiency to reduce emissions to air. Our manufacturing units are governed by applicable environmental permits, which set limits for emissions to air. Our most significant emissions to air are carbon dioxide (CO_2) and volatile organic compounds (VOC).

Our Scope 1 emissions essentially stayed flat in 2018 compared to 2017, in line with our energy use. Scope 1 emissions cover methane (CH_4) and nitrous oxide (N_2O) gases besides CO_2. Emissions are calculated based on reported fuel consumption and emission factors according to GHG Protocol/IEA 2018.

Our Scope 2 emissions increased by less than 2%. The emissions are calculated according to a location-based method, based on reported secondary energy consumption and emission factors. Emissions factors are calculated according to GHG Protocol/IEA 2018.

Our GHG intensity per sellable ton produced decreased by 1%. Given the reduced energy intensity, we have reduced the GHG emissions by 8,500 metric tons of CO_2 eqv. compared to 2017 when corrected for the production volume. The emission intensity and reduction cover Scope 1 and Scope 2 emissions.

More than 70% of total greenhouse gas emissions generated by our operations originate from purchased electricity. The indirect CO_2 emissions are highly dependent on the mix of the energy sources available in the operating countries' national energy grids. A project to investigate options to increase renewable energy will start in 2019.

Our emissions of Volatile Organic Compounds (VOC) are mainly related to solvent-based printing inks. Our manufacturing units with recognizable VOC emissions typically have a VOC incinerator or a VOC recovery unit installed. The recovery system enables recycling and reuse of the solvent used in printing inks. We also use water-based printing inks, which are solvent free and do not result in VOC emissions (although they require greater use of energy for evaporation). In 2018, 16% (16%) of printing inks used in the Group were water- or oil-based.

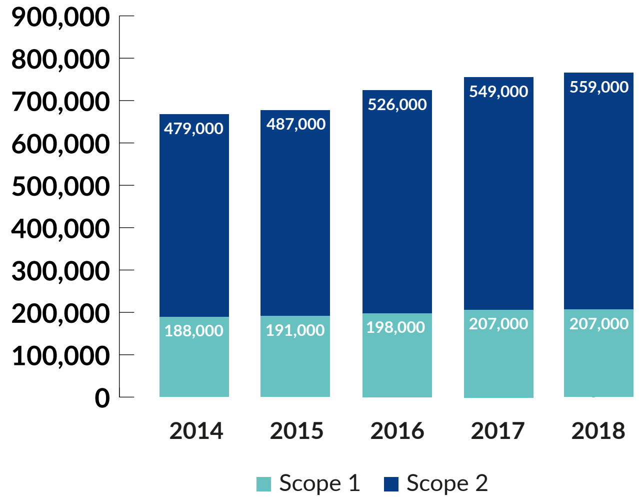
Waste

306-2 Waste by type and disposal method

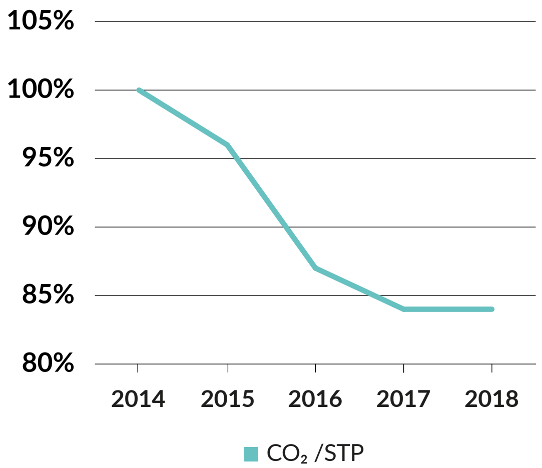
The main waste streams in our processes are cutting waste from paperboard packaging manufacturing, materials rejected during the fiber pulping process, and plastic waste. Cutting waste from our paperboard packaging manufacturing is either recycled as raw material for fiber packaging inside the Group or sent for external recycling. In general, material recovery refers to either internal or external recycling. The waste disposal method is determined on the manufacturing unit level based on the available infrastructure and type of waste.

Our total amount of waste increased 4%, mainly due to growth in our Foodservice and North America segments. The amount of hazardous waste represented 4% (4%) of the total waste amount. Hazardous waste was treated locally by dedicated hazardous waste handlers in line with local regulatory requirements.

Total greenhouse gas emissions (metric tons CO_2 eqv)



Indexed greenhouse gas emissions (metric tons CO_2 eqv)



Waste by type and disposal method (metric tons)

		2018	2017	2016	2015	2014
Non-hazardous waste	Landfill	20,000	21,000	18,000	18,000	19,000
	Recycling	157,000	149,000	131,000	113,000	107,000
	Energy recovery	13,000	13,000	16,000	16,000	13,000
Hazardous waste	Landfill	1,200	900	1,200	700	500
	Recycling	4,500	4,200	4,000	2,800	3,000
	Energy recovery	1,100	1,100	700	700	1,000
Total		196,800	189,200	170,900	151,200	143,500
Recycling rate (%)		82	81	79	76	77
Hazardous waste (% of total)		4	4	3	3	3

Incineration without energy recovery is applied to less than 1% of total waste and therefore left out of this table for the sake of clarity.

Economic responsibility

As a publicly listed company, Huhtamaki is committed to earning profit for its shareholders. Huhtamaki has several financial long-term ambitions, which are available in our investor relations materials. The most important of these are our growth ambition, 5% organic growth complemented by 5% growth by acquisitions, and our profitability ambition of 10+% EBIT margin. In addition, our ambition is to pay 40–50% of our profit as dividend to our shareholders.

Huhtamaki's economic responsibility in 2018

In 2018, the economic value generated by the Group was EUR 1,049 million (EUR 1,063 million) of which EUR 786 million (EUR 760 million) was distributed to stakeholders and EUR 263 million (EUR 303 million) retained in the Group for operational development and further growth. The Group achieved 5% organic and 3% acquisitive growth, and a 7.2% EBIT margin. (Adjusted EBIT margin, excluding IAC of EUR -26 million, was 8.0%.) Dividends paid to shareholders correspond to a 42% payout ratio (of 2017 profit).

Huhtamaki as a tax payer

Huhtamaki is committed to paying all taxes and complying with other related obligations according to local laws and regulations. Moreover, the Huhtamaki values, the requirements of our Code of Conduct and the Code of Conduct for Huhtamaki Suppliers transcend national boundaries and complement the framework for our business operations. Huhtamaki's Public Tax Strategy is available on our website.

One of our operating principles is to ensure the predictability and optimization of taxation. In addition to ensuring that taxes are paid correctly in all locations, we also seek to ensure that we do not pay excess taxes and that we capitalize on tax deductions enabled by local tax regulations. The Group's tax expenses in 2018 decreased to EUR 38 million (EUR 50 million) and paid taxes amounted to EUR 38 million (EUR 44 million). The corresponding tax rate was 19% (20%).

In intercompany business transactions, we comply with the OECD guidelines on transfer pricing. Intersegment net sales were EUR 16 million (EUR 17 million), less than 1% of the Group's net sales. Our business decisions are made keeping our strategy, financial ambitions and commercial environment in mind whilst at the same time aiming to serve our customers better. While taxation is one factor to be considered, it is not a dominant factor.

GRI 201-1 Direct economic value generated and distributed (MEUR)

		2018	2017
Customers	Net sales	3,104	2,989
Suppliers	Purchases	2,055	1,926
	Economic value generated	1,049	1,063
Personnel	Personnel expenses	637	616
Shareholders	Dividends	83	76
Debt investors	Net financial expenses	27	18
Public sector	Taxes	38	50
	Economic value distributed	785	760
	Economic value retained in the Group	264	303

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Huhtamaki's own KPI	All plants in water stressed areas have a water management plan	47	6, 12, 14
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305-2	Energy indirect (Scope 2) GHG emissions	47, 48	9, 12, 13, 14, 15
305-4	GHG emissions intensity	47, 48	9, 12, 13, 14, 15
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Huhtamaki's own KPI	Community programs in place on site level and Group level	40	17
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Huhtamaki's own KPI	Supply chain due diligence system in place covering ethical, social and environmental criteria	44	5, 8
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